

6.3 Carry chains in FPGAs

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■ Create an *adder* w/ FPGAs:

$$s_i = a_i \oplus b_i \oplus c_i$$

$$c_{i+1} = a_i b_i + a_i c_i + b_i c_i$$

- The most naïve method: use FPGA logic blocks to generate the *sum* and *carry* for each bit
 - A LUT4 can generate the *sum*, and another LUT4 will be required to realize the *carry* equation.
 - Carry output from each bit needs to be forwarded to the next bit using interconnect resources.
- Many FPGAs provide dedicated circuitry for **generating** and **propagating carry bits** to subsequent higher bits (*dedicated carry chain*).

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